EXACT SALES & NON-SALES INCENTIVE

IDSPL Incentive

Exact sales payouts. Non-sales allocations.
Understand both incentive reports.

1

EXACT SALES INCENTIVE

Total Incentive
× Applied Percentage
= Exact Sales Payout

2

NON-SALES INCENTIVE

Outlet IDS Fund
× Department Allocation Rate
= Non-sales Team Amount

STEP 1 **Upload sales** → STEP 2 **Calculate** → STEP 3 **Check reports** → STEP 4 **Download**

Two reports. Clear calculations.

- Sales: outlet and category group payouts after applied share
- Non-sales: Accounts, Ware House, Customer Care, Support & HO
- Select SALES, NON SALES or BOTH; download Excel or PDF

Illustrations explain the workflow, not live payout amounts.
Calculation reference reviewed on 8 October 2026.

Incentive Project

Detailed backend calculation sheets | Initiative ERP / IDSPL

A source-based reference for every incentive report sheet, rate, formula, rounding step and export dependency.

Calculation pipeline

> **UPLOAD → PARSE → NORMALIZE → AGGREGATE → CALCULATE → EXPORT**

Accepted inputs are .xlsx, .xlsm and text-based PDF. The backend accepts ALM, ASH, HZT, GNG and VKN for incentive calculations. Editable inputs default to a 7% profit rate and 2.5% incentive rate.

All seven sheets in the full workbook

Sheet	Backend result / purpose
Outlet Summary	summary: basic incentive per outlet
Category Incentive	category_summary: HA, HE, MOB, COM, DC, ACC
Grouped Incentive	grouped_summary: each normalized outlet/category group
Sales Incentive	exact_summary: payout after outlet/group applied share
IDS Fund	ids_fund_report.rows: category incentive and fund contribution
IDS Fund Outlet Summary	ids_fund_report.summary: fund total per outlet
Non-sales Staff Incentive	Allocation percentages, calculated amounts and Central Service totals

Two calculation branches

Sales payout branch: sales × profit rate × incentive rate × applied share. Intermediate amounts are rounded.

IDS Fund branch: category sales × fixed category/outlet rate × fund share. This fund then supplies non-sales allocations. The editable 7% and 2.5% inputs do not drive IDS Fund rates.

> The selected-export workbook contains Sales Incentive and/or Non-sales Incentive as calculated values. It is different from the seven-sheet full workbook. This document describes current code, not a new payout policy.

Reviewed: 8 October 2026. Source: main.py and static/incentive.html. Examples use synthetic data, not live business records.

Input & Aggregation

01 / How source rows become calculation inputs

Normalization rules

Input	Backend behaviour
Outlet names	Alambagh → ALM; Ashiyana → ASH; Hazratganj / HTZ → HZT; Gomtinagar → GNG; Vikas Nagar → VKN
Categories	Recognizes HA, HE, MOB, COM, DC, ACC and ACC COM. ACC COM is identified before COM / ACC.
Amounts	Numeric values and numeric text are accepted; commas / currency text are removed. Parenthesized numbers become negative.
Rate parsing	Values from 0 through 1 are multiplied by 100; e.g. 0.85 → 85. This parser also interprets numeric 1 as 100.
Workbook formulas	Read with data_only=True: saved cached values are used, not recalculated formulas.
Duplicates	Identical parsed rows are deduplicated within each sheet using month, outlet, category, sales, applied rate and exact amount. No global cross-sheet deduplication.

Summary rows and recognized outlets

The parser prefers detail rows over rows identified as ALL / OTHER outlets or ALL / TOTAL / GRAND TOTAL categories. If no detail exists, it retains the original rows. Calculations then ignore outlets outside the five supported codes.

> **outlet_totals[outlet] += sales**
> **grouped_totals[outlet, category] += sales**
> **category_totals[outlet, category] += sales only for the six core categories**

ACC COM contributes to outlet and grouped totals, and to the fund and exact sales rules. It is excluded from the six-category Category Incentive sheet. An unrecognized category becomes ALL in the basic calculations; the fund branch requires a recognized category.

Period and imported payout fields

Month is parsed but is not an aggregation key in calculate_incentive_report. Multiple months in one file are combined. Use one intended reporting period per upload.

Imported applied_rate and exact_incentive fields can be parsed, but build_sales_incentive_summary uses the hard-coded rules to recalculate exact payouts. Imported payout values do not override those rules.

PDF parsing uses extracted text and the last numeric value on a matching line as sales. Review the resulting totals carefully when source lines contain several amounts. Scanned-image OCR is not implemented in this incentive parser.

Basic Calculation Sheets

02 / Outlet Summary, Category Incentive and Grouped Incentive

Shared formula: _incentive_calc

> **P = round(S × profit_rate / 100, 2)**
> **I = round(P × incentive_rate / 100, 2)**
> **Returned sales = round(S, 2)**

S is aggregated sales, P is estimated average profit and I is the basic total incentive. Profit is rounded before the second multiplication. Backend calculations use Python floats and round.

Sheet	Grouping	Export columns
Outlet Summary	One row per supported outlet	Outlet, Total Sales, AVG Profit, Total Incentive
Category Incentive	One row for each outlet and each of six categories	Outlet, Category, AVG Profit, Total Incentive
Grouped Incentive	One row per outlet and normalized category/group	Outlet, Category Group, Total Sales, AVG Profit, Total Incentive

Worked example at default rates

Sales S	Average profit P at 7%	Basic incentive I at 2.5%
100,000.00	7,000.00	175.00
200,000.00	14,000.00	350.00
400,000.00	28,000.00	700.00
700,000.00	49,000.00	1,225.00

Totals and reconciliation

The API grand total applies _incentive_calc to the sum of all supported outlet sales. It does not sum the individually rounded outlet incentives, so small rounding differences are possible.

The category API total recalculates from sales in the six core categories. The exported Category Incentive sheet writes category rows, including zero rows; it does not write the API category-total row.

With Rs. 100,000 in each of seven categories including ACC COM, outlet sales are Rs. 700,000 and the basic incentive is Rs. 1,225. The six-category total is Rs. 600,000 and Rs. 1,050. The Rs. 175 difference is ACC COM.

Stored values, rather than Excel formulas
These three sheets export backend-calculated numbers. Changing sales in the downloaded workbook does not automatically recalculate their profit and incentive columns.

Sales Incentive Sheet

03 / Fixed payout shares by outlet and category group

> **Group sales S = sum of eligible category sales**
> **Base I = round(round(S × profit_rate / 100, 2) × incentive_rate / 100, 2)**
> **Exact payout = round(I × applied_rate / 100, 2)**

Outlet	Eligible category group	Applied share
ALM	HA + HE	85%
ALM	MOB + DC + ACC + ACC COM	13%
ALM	COM	78%
ASH	ALL normalized categories	100%
GNG	ALL normalized categories	52%
HZT	HA + HE	65%
HZT	MOB + COM + DC + ACC + ACC COM	60%
VKN	ALL normalized categories	55%

The displayed ALM group label is MOB + DC + ACC and HZT label is MOB + COM + DC + ACC; their category sets also include ACC COM. A group with zero total sales is omitted. Nonzero negative totals are not omitted by this check.

Worked example: Rs. 100,000 in each of seven categories per outlet

Outlet / group	Sales	Base incentive	Exact payout
ALM HA + HE	200,000	350.00	297.50
ALM MOB / DC / accessories	400,000	700.00	91.00
ALM COM	100,000	175.00	136.50
ASH ALL	700,000	1,225.00	1,225.00
GNG ALL	700,000	1,225.00	637.00
HZT HA + HE	200,000	350.00	227.50
HZT mobile / IT / accessories	500,000	875.00	525.00
VKN ALL	700,000	1,225.00	673.75

Sum of example exact payouts: Rs. 3,813.75. The Sales Incentive sheet writes Outlet, Category Group, Total Incentive, Applied Rate and Exact Incentive as values. The applied rate is stored as a fractional number, e.g. 85% as 0.85.

IDS Fund Calculation

04 / Category incentive rates and fund contributions

> **Category incentive = round(category sales × category rate / 100, 2)**
> **IDS Fund = round(category incentive × fund share / 100, 2)**

Category	Rate: ALM / ASH / GNG / VKN	Rate: HZT	Fund share
MOB	0.125%	0.125%	30%
COM	0.25%	0.25%	20%
DC	0.25%	0.25%	25%
HA	0.25%	0.25%	20%
HE	0.25%	0.25%	25%
ACC	1%	0.5%	25%
ACC COM	1%	0.25%	25%

Rs. 100,000 sales in each category

Category	Non-HZT incentive / fund	HZT incentive / fund
MOB	125.00 / 37.50	125.00 / 37.50
COM	250.00 / 50.00	250.00 / 50.00
DC	250.00 / 62.50	250.00 / 62.50
HA	250.00 / 50.00	250.00 / 50.00
HE	250.00 / 62.50	250.00 / 62.50
ACC	1,000.00 / 250.00	500.00 / 125.00
ACC COM	1,000.00 / 250.00	250.00 / 62.50
Total per outlet	3,125.00 / 762.50	1,875.00 / 450.00

Backend totals

Category amounts are rounded first, then summed by outlet. All five example outlets together produce Rs. 3,500,000 sales, Rs. 14,375 category incentive and **Rs. 3,500 IDS Fund**.

If a retained fund row has no rate/share, its incentive and fund become null; that outlet's related totals become null, and grand incentive/fund totals become null if any row is pending. If source rows exist but no fund rows can be built, insufficient_data is set.

Fund Sheet Formulas

05 / IDS Fund and IDS Fund Outlet Summary in Excel

IDS Fund cell map

Data starts on row 5. Columns: A Outlet; B Category; C Total Sales; D Incentive Rate; E Total Incentive; F Fund Share; G IDS Fund. D and F store fractions, such as 0.00125 and 0.30.

```
E5 = IF(ISNUMBER(D5),ROUND(C5*D5,2),"Pending")
G5 = IF(AND(ISNUMBER(E5),ISNUMBER(F5)),
  ROUND(E5*F5,2),"Pending")
```

The formulas repeat for every data row. The grand-total sales cell uses SUM. Incentive and fund grand totals require all cells in the respective range to be numeric:

```
=IF(COUNT(E5:E[last])=ROWS(E5:E[last]),
  SUM(E5:E[last]),"Pending")
```

[last] represents the final data row; actual exports use its numeric row number. The fund total uses the same formula pattern with column G.

IDS Fund Outlet Summary

Row 1 is the title, row 2 the header, and data begins at row 3. For each outlet, column B adds its matching IDS Fund G cells directly. Example: `='IDS Fund'!G5+'IDS Fund'!G6`. The final row uses SUM over outlet amounts.

Calculation behaviour to understand

Direct addition of a text Pending value in the outlet-summary formula can produce an Excel error instead of Pending. Its final SUM may ignore text or propagate referenced errors; it is not the same all-numeric guard as the IDS Fund grand total.

OpenPyXL writes formulas but does not calculate them. A spreadsheet application must recalculate the downloaded workbook. Other basic and exact sheets retain the values originally computed by the backend.

Sheet dependency
Source sales and fund rates → IDS Fund formulas → outlet fund formulas → full-workbook non-sales formulas. Editing a fund input may update these linked sheets after recalculation, but does not update the basic or exact sales sheets.

Formula examples are transcribed from build_incentive_workbook in main.py. The empty fund-data case creates an inverted total range; verify a workbook with no recognized fund rows before using it.

Non-sales Allocations

06 / Accounts, Ware House, Customer Care, Support and HO

Allocation rules

Location	Accounts	Customer Care	Support	HO
ALM	35%	12%	15%	15%
ASH	35%	12%	15%	10%
HZT	35%	12%	15%	12%
GNG	35%	12%	15%	10%
VKN	35%	12%	15%	10%

Outlet allocation = round(outlet IDS Fund × allocation / 100)
Ware House = round(sum of available outlet funds × 16%, 2)

Accounts, Customer Care, Support and HO are rounded to whole rupees. Ware House is a pooled allocation rounded to two decimals, shown on a separate MWH row. It is not 16% separately calculated and rounded for each outlet.

Worked example using funds from page 5

Location	Fund	Accounts	Customer Care	Support	HO
ALM	762.50	267	92	114	114
ASH	762.50	267	92	114	76
HZT	450.00	158	54	68	54
GNG	762.50	267	92	114	76
VKN	762.50	267	92	114	76
Column totals	3,500.00	1,226	422	524	396

Ware House: Rs. 3,500 × 16% = Rs. 560.00. These example allocations total Rs. 3,128.00. The code does not automatically allocate the remaining fund balance to any department.

Missing funds and rounding

A missing outlet fund yields null values for that outlet. The backend warehouse pool uses available funds even when some outlets are missing; if none exist, the pool and warehouse amount are null. Department totals sum available numeric allocations and can therefore be partial.

Python round uses ties-to-even for exact halfway cases; Excel ROUND uses a different tie rule. For example, Python round(2.5) = 2, whereas Excel ROUND(2.5,0) = 3. Binary floating-point representation can also influence rounding. Selected exports use backend results; the full workbook recalculates formulas in Excel.

Exports & Review Checks

07 / Non-sales cell map, export variants and implementation limits

Full workbook: Non-sales Staff Incentive

A Location; B IDS Fund; C Accounts; D Ware House; E Customer Care; F Support; G HO. Rows 2-6 contain outlet fund links and percentage inputs. Row 7 is MWH with 16% in D7.

```
B7 = IF(COUNT(B2:B6)=5,SUM(B2:B6),"Pending")
C11 = IF(ISNUMBER(B2),ROUND(B2*C2,0),"Pending")
D16 = IF(ISNUMBER(B7),ROUND(B7*D7,2),"Pending")
```

Rows 11-15 contain outlet allocation amounts. E, F and G use the same pattern as C with their respective rate columns. Row 16 holds the warehouse calculation. Row 18 is Central Service; HO subtotal is first written to G17, then referenced by G18.

Verified linking limitation

Positional links can assign the wrong fund
The fund outlet summary is sorted ALM, ASH, GNG, HZT, VKN when all five exist. The staff sheet labels rows ALM, ASH, HZT, GNG, VKN and links by row position. HZT and GNG therefore receive each other's funds in the full workbook. Missing outlets can shift more links. This document reports the issue; it does not modify the application.

The backend non-sales calculation maps funds by outlet name. Selected exports use these backend amounts. The full workbook warehouse formula requires all five linked funds, while the backend selected report pools whatever funds are available. Full-workbook department SUM formulas can omit Pending text and show partial totals.

Export variants

Endpoint	Output
/api/incentive/export	Full seven-sheet Outlet_Incentive_Report.xlsx
/api/incentive/selected-export	sales, non-sales or both; Excel or PDF
/api/incentive/exact-export	Sales Incentive only; Excel or PDF

Review checklist

Use one reporting period; reconcile source sales and supported outlets; account for ACC COM; distinguish base incentive, exact payout and fund; recalculate Excel; verify outlet links, missing funds and rounding before approving amounts.

Primary backend functions: parse_incentive_upload, calculate_incentive_report, _incentive_calc, build_sales_incentive_summary, build_ids_fund_report, build_non_sales_report, build_incentive_workbook and selected export builders.